Lecture - 12

$(V, \oplus, \odot, \mathbb{F})$

Inner product,

$$\boxed{B(u,v) \longrightarrow \mathbb{F}}$$

$\langle u,v \rangle$ angle bracket

(i) $\langle u,v \rangle = \langle v,u \rangle$

(ii) $\langle \alpha u,v \rangle = \alpha \langle u,v \rangle$

(iii) $\langle u,u \rangle > 0$ except $u=0$,

$\langle u,u \rangle = 0$

defined by the Teacher.

$$\left\langle \begin{pmatrix} a \\ b \end{pmatrix}, \begin{pmatrix} c \\ d \end{pmatrix} \right\rangle = ac + bd$$

$$\langle f(x), g(x) \rangle = \int_{-\pi}^{\pi} f(x)\, g(x)\, dx$$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} = 2+6 = 8$$

$$f\left(\underset{\substack{\cap \\ V}}{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}, \underset{\substack{\cap \\ V}}{\begin{bmatrix} 2 \\ 3 \end{bmatrix}}\right) = 8$$

(i) $f(u,v) = f(v,u)$

(ii) $f(\alpha u, v) = \alpha f(u,v)$

(ii)

$$\alpha \begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} \alpha \\ 2\alpha \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$= 2\alpha + 6\alpha = 8\alpha$$

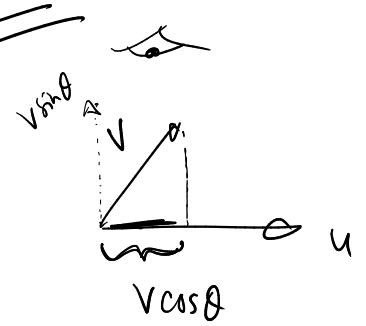
$$\alpha \left(\begin{bmatrix} 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right) = \alpha \times 8 = 8\alpha$$

(iii)

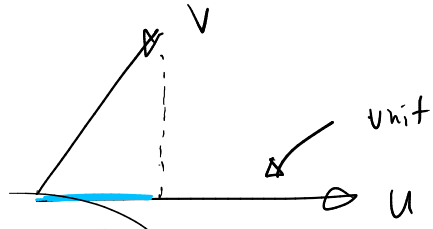
$$f(u,u) > 0$$

$$\begin{bmatrix} -1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = 5 > 0$$

Dot product



$$u \cdot v = |u| \underbrace{|v| \cos\theta}_{\text{Proj}_u(v)}$$



unit vector. $\hat{u} = \dfrac{u}{|u|}$

$$|v| \cos\theta = \text{Proj}_u(v)$$

this vector is projected

we are projecting in this direction

$$\boxed{\text{Proj}_u(v) = \frac{u \cdot v}{|u|}}$$

Value.

$$= \frac{\langle u, v \rangle}{\sqrt{\langle u, u \rangle}}$$

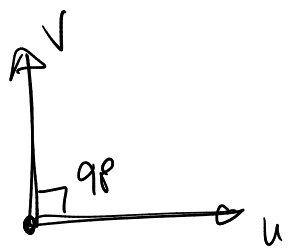
$$\overrightarrow{\text{Proj}_u(v)} = \frac{\langle u, v \rangle}{\sqrt{\langle u, u \rangle}} \times \frac{u}{|u|}$$

$$= \frac{\langle u, v \rangle}{\sqrt{\langle u, u \rangle}} \frac{u}{\sqrt{\langle u, u \rangle}}$$

$$\boxed{\overrightarrow{\text{Proj}_u(v)} = \frac{\langle u, v \rangle}{\langle u, u \rangle} u}$$

✓ ★★★

Orthogonal & orthonormal



$u \cdot v = 0$

$\langle u, v \rangle = 0$

$u \cdot v = 0$ or $\langle u, v \rangle = 0$

$\underbrace{|u| = 1}_{\text{unit}}$, $\underbrace{|v| = 1}_{\text{unit}}$

Example

in	out
x	x
-2	-2
-1	-1
0	0
1	1
2	2

x	x^2
-2	4
-1	1
0	0
1	1
2	4

$$\langle P(x), q(x) \rangle = \sum_{k=1}^{5} P(x_k)\, q(x_k) = P(-2)q(-2) + P(-1)q(-1) + P(0)q(0) + P(1)q(1) + P(2)q(2)$$

$$\langle x, x^2 \rangle = -2\times 4 + (-1)\times 1 + 0\times 0 + 1\times 1 + 2\times 4$$

$$= 0.$$

Becarse, the inner product is zero.

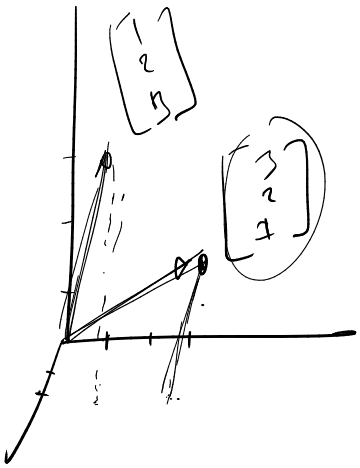
Example

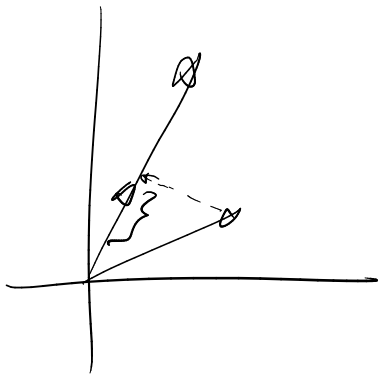
$$S = \{1, \sin(x), \cos(x)\}$$

$$\langle \sin(x), \cos(x) \rangle = \frac{1}{2}\int_{-\pi}^{\pi} 2\sin(x)\cos(x)\, dx$$

$$= \frac{1}{2}\int_{-\pi}^{\pi} \sin(2x)\, dx = \frac{1}{2}\left(\frac{-\cos(2x)}{2}\Big|_{-\pi}^{\pi} \right) = 0.$$

Example





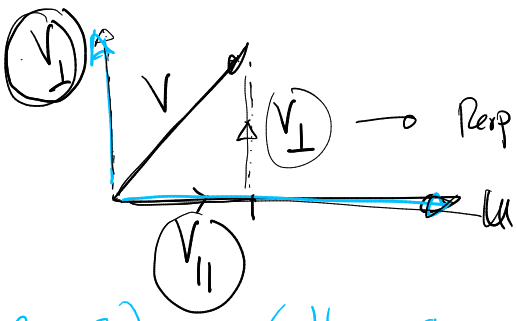
$$\mathrm{Proj}_{(1,2,3)}(3,2,1) = \frac{\langle (3,2,1),(1,2,3)\rangle}{\langle (1,2,3),(1,2,3)\rangle}(1,2,3)$$

$$= \frac{3+4+3}{1+4+9}(1,2,3)$$

$$= \frac{10}{14}(1,2,3) = \left(\frac{10}{14}, \frac{20}{14}, \frac{30}{14}\right)$$

$u = (1,2,3)$ & $v = (3,2,1)$

$$V_{\parallel} + V_{\perp} = V$$



$$V_{\perp} = V - V_{\parallel}$$

$$= V - \mathrm{Proj}_u(V)$$

$$V_{\perp} = (3,2,1) - \left(\frac{10}{14}, \frac{20}{14}, \frac{30}{14}\right) = \left(\frac{16}{7}, \frac{4}{7}, -\frac{8}{7}\right)$$

Example

$$\underbrace{v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}}_{u_1} \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \qquad u_2 = v_2 - \underbrace{\mathrm{Proj}_{u_1}(v_2)}$$

$$= v_2 - \frac{\langle u_1, v_2 \rangle}{\langle u_1, u_1 \rangle} u_1$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} - \frac{0}{*} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$u_3 = v_3 - \mathrm{Proj}_{u_1}(v_3) - \mathrm{Proj}_{u_2}(v_3)$$

$$= \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{\left\langle \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix} \right\rangle}{\left\langle \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} \right\rangle} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix} - \frac{\cancel{2}}{\cancel{4}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{-\cancel{4}}{\cancel{4}} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -1 \\ 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \\ 0.5 \\ -0.5 \end{bmatrix}$$

orthogonal vectors $\boxed{u_1, u_2, u_3}$

Orthonormal vectors, $\tilde{u}_1 = \frac{u_1}{|u_1|}$, $\tilde{u}_2 = \frac{u_2}{|u_2|}$, $\tilde{u}_3 = \frac{u_3}{|u_3|}$

$$= \frac{u_1}{\sqrt{\langle u_1, u_1 \rangle}}, \quad = \frac{u_2}{\sqrt{\langle u_2, u_2 \rangle}} \quad \frac{u_3}{\sqrt{\langle u_3, u_3 \rangle}}$$

$$\boxed{\text{Any independent vectors}} \xrightarrow[\text{Process}]{\text{Gram-Schmidt}} \boxed{\text{Orthonormal vectors}}$$

$$V_1 = 1, \quad V_2 = x, \quad V_3 = x^2$$

$$\boxed{\langle f, g \rangle = \int_{-\pi}^{\pi} f(x) g(x)\, dx}$$

$$u_1 = V_1 \qquad u_2 = V_2 - \frac{\langle u_1, V_2 \rangle}{\langle u_1, u_1 \rangle} u_1 \qquad u_3 = V_3 - \frac{\langle u_1, V_3 \rangle}{\langle u_1, u_1 \rangle} u_1 - \frac{\langle u_2, V_3 \rangle}{\langle u_2, u_2 \rangle} u_2$$